

1. Write the program to create a file filename.txt and perform the following operations.
 - a. Retrieve the file name
 - b. Retrieve the file absolute path
 - c. Check if the txt file has write and read permissions

```
//Question Number 1

import java.io.File;
import java.io.IOException;

public class FileDetails {
    public static void main(String[] args) {
        File file = new File("filename.txt");

        try {
            // Create the file if it doesn't exist
            if (file.createNewFile()) {
                System.out.println("File created successfully.");
            } else {
                System.out.println("File already exists.");
            }

            // Retrieve and print file details
            System.out.println("File Name: " + file.getName());
            System.out.println("Absolute Path: " +
file.getAbsolutePath());
            System.out.println("Can Read: " + file.canRead());
            System.out.println("Can Write: " + file.canWrite());

        } catch (IOException e) {
            System.out.println("An error occurred: " + e.getMessage());
        }
    }
}
```

```

Cannot load PSReadline module. Console is running without PSReadline.
PS E:\Jaca\Workshop\INE> & 'C:\Program Files\Eclipse Adoptium\jdk-21.0.7.6-hotspot\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\Vajju Shrestha\AppData\Roaming\Code\User\workspaceStorage\c328cf7201fe0fff702f6e29bc14c3af\redhat-.java\jdt_ws\Wor
kshop\INE_4cd97d9\bin' 'FileDetails'
File created successfully.
File Name: filename.txt
Absolute Path: E:\Jaca\Workshop\INE\filename.txt
Can Read: true
Can Write: true
PS E:\Jaca\Workshop\INE>

```

2. Write a program to check if the file or directory specified by the path name exists or not.

```

//Question number 2

import java.io.File;

public class CheckExistence {
    public static void main(String[] args) {
        // Specify the path to check
        String path = "example.txt"; // Change this to any file or
directory path
        File file = new File(path);

        if (file.exists()) {
            System.out.println("The file/directory exists.");
        } else {
            System.out.println("The file/directory does not exist.");
        }
    }
}

```

```

Cannot load PSReadline module. Console is running without PSReadline.
PS E:\Jaca\Workshop\INE> & 'C:\Program Files\Eclipse Adoptium\jdk-21.0.7.6-hotspot\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\Vajju Shrestha\AppData\Roaming\Code\User\workspaceStorage\c328cf7201fe0fff702f6e29bc14c3af\redhat-.java\jdt_ws\Wor
kshop\INE_4cd97d9\bin' 'CheckExistence'
The file/directory does not exist.
PS E:\Jaca\Workshop\INE>

```

3. List out the differences between byte-oriented and character-oriented streams.

Feature	Byte-Oriented Streams (InputStream , OutputStream)	Character-Oriented Streams (Reader , Writer)
Data Type	Handles binary data (bytes)	Handles text data (characters)
Unit of Data	1 byte (8 bits)	2 bytes (16-bit Unicode characters)
Encoding Awareness	Not aware of character encoding	Encoding-aware (uses character encoding)
Suitable For	Images, audio, videos, PDFs, etc.	Text files (like .txt , .csv)
Base Classes	InputStream , OutputStream	Reader , Writer
Examples	FileInputStream , BufferedInputStream	FileReader , BufferedReader
Use Case	When dealing with raw binary files	When working with readable character data

4. Write a program to read the contents of the file line by line using `BufferInputStream` and `BufferedReader`.

```
import java.io.*;

public class ReadFileBothWays {
    public static void main(String[] args) {
        String filePath = "sample.txt";
```

```
// Using BufferedReader (character stream)
System.out.println("Reading with BufferedReader:");
try (BufferedReader br = new BufferedReader(new
FileReader(filePath))) {
    String line;
    while ((line = br.readLine()) != null) {
        System.out.println(line);
    }
} catch (IOException e) {
    System.out.println("Error using BufferedReader: " +
e.getMessage());
}

// Using BufferedInputStream (byte stream)
System.out.println("\nReading with BufferedInputStream:");
try (BufferedInputStream bis = new BufferedInputStream(new
FileInputStream(filePath))) {
    StringBuilder line = new StringBuilder();
    int ch;
    while ((ch = bis.read()) != -1) {
        if (ch == '\n') {
            System.out.println(line.toString());
            line.setLength(0); // clear buffer
        } else if (ch != '\r') {
            line.append((char) ch);
        }
    }
    // Print last line if it doesn't end with a newline
    if (line.length() > 0) {
        System.out.println(line);
    }
} catch (IOException e) {
    System.out.println("Error using BufferedInputStream: " +
e.getMessage());
}
}
```

5. Write the program to read the content of one.txt file and write the content to the two.txt file using `BufferInputStream` and `BufferOutputStream`.

```
import java.io.*;

public class ReadFileBothWays {
    public static void main(String[] args) {
        String filePath = "filename.txt";

        // Using BufferedReader (character stream)
        System.out.println("Reading with BufferedReader:");
        try (BufferedReader br = new BufferedReader(new
FileReader(filePath))) {
            String line;
            while ((line = br.readLine()) != null) {
                System.out.println(line);
            }
        } catch (IOException e) {
            System.out.println("Error using BufferedReader: " +
e.getMessage());
        }

        // Using BufferedInputStream (byte stream)
        System.out.println("\nReading with BufferedInputStream:");
        try (BufferedInputStream bis = new BufferedInputStream(new
FileInputStream(filePath))) {
            StringBuilder line = new StringBuilder();
            int ch;
            while ((ch = bis.read()) != -1) {
                if (ch == '\n') {
                    System.out.println(line.toString());
                    line.setLength(0); // clear buffer
                } else if (ch != '\r') {
                    line.append((char) ch);
                }
            }
            // Print last line if it doesn't end with a newline
        }
    }
}
```

```

        if (line.length() > 0) {
            System.out.println(line);
        }
    } catch (IOException e) {
        System.out.println("Error using BufferedInputStream: " +
e.getMessage());
    }
}
}
}

```

```

PS E:\Java\Workshop\WNE>
PS E:\Java\Workshop\WNE> cd 'e:\Java\Workshop\WNE' & 'C:\Program Files\Eclipse Adoptium\jdk-21.0.7.6-hotspot\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\Vajju Shrestha\AppData\Roaming\Code\User\workspaceStorage\c320cf7281fe8fff702f6c29
bc14c3d9\redhat-java\bin\java\bin' 'ReadFileWithWays'
Reading with BufferedReader:
hello this is a file
Reading with BufferedInputStream:
hello this is a file
PS E:\Java\Workshop\WNE>

```

6. Rewrite the program from Question no 5 using BufferedReader and BufferedWriter.

```

import java.io.*;

public class CheckExistenceBuffered {
    public static void main(String[] args) {
        String inputFile = "filename.txt"; // Contains the path to
check
        String outputFile = "output.txt"; // Will store the result

        try (
            BufferedReader reader = new BufferedReader(new
FileReader(inputFile));
            BufferedWriter writer = new BufferedWriter(new
FileWriter(outputFile))
        ) {
            String path = reader.readLine(); // Read path from file
            File file = new File(path);

            if (file.exists()) {
                writer.write("The file/directory exists.");
            }
        }
    }
}

```

```
        } else {  
            writer.write("The file/directory does not exist.");  
        }  
  
        System.out.println("Check completed. Result written to  
output.txt");  
  
    } catch (IOException e) {  
        System.out.println("An error occurred: " + e.getMessage());  
    }  
}  
}
```

```
PS E:\Java\Workshop\IDE>  
PS E:\Java\Workshop\IDE> cd "e:\Java\Workshop\IDE"; & "C:\Program Files\Eclipse Adoptium\jdk-21.0.7-hotspot\bin\java.exe" -XX:+ShowCodeDetailsInExceptionMessages -cp "C:\Users\Vajju Shrestha\AppData\Roaming\Code\User\workspaceStorage\c320cf72-01f8bfff782f6298c14c3a1\redhat-java\jdk_vs_Workshop\IDE_f4cd97d9\bin" "CheckExistenceBuffered"  
Check completed. Result written to output.txt  
PS E:\Java\Workshop\IDE> []
```

7. Write the program to append the text into the existing text file.

```
import java.io.*;  
  
public class AppendToFile {  
    public static void main(String[] args) {  
        String filePath = "filename.txt"; // Target file  
        String textToAppend = "This text will be appended to the file.\n";  
  
        try (BufferedWriter writer = new BufferedWriter(new  
FileWriter(filePath, true))) {  
            writer.write(textToAppend);  
            System.out.println("Text appended successfully.");  
        } catch (IOException e) {  
            System.out.println("An error occurred: " + e.getMessage());  
        }  
    }  
}
```

```
Cannot load PSReadline module. Console is running without PSReadline.  
PS E:\Java\Workshop\INE> & "C:\Program Files\Eclipse Adoptium\jdk-21.0.7.6-hotspot\bin\java.exe" "-XX:+ShowCodeDetailsInExceptionMessages" "-cp" "C:\Users\Vejju Shrestha\AppData\Roaming\Code\User\workspaceStorage\c320cf7201fe0ff7f02fe29bc14c3af\redhat  
.java\jdk_ws\Workshop\INE_44cd97d9\bin" "AppendToFile"  
Text appended successfully.  
PS E:\Java\Workshop\INE>
```

8. You have the text file with the given number range from 1 to 10. Each number is separated by space in between. Write the program to read the number from the text file then write the odd number into odd.txt file and even number into even.txt file. The program should have the following methods.

- `public List<Integer> getEven(List<Integer> numbers);`
- `public List<Integer> getOdd(List<Integer> numbers);`
- `public void write(List<Integer> numbers, String filePath);`
- `public void read(String filePath);`

```
import java.io.*;  
import java.util.*;  
  
public class NumberSeparator {  
  
    public static void main(String[] args) {  
        NumberSeparator ns = new NumberSeparator();  
        ns.read("filename.txt");  
    }  
  
    public void read(String filePath) {  
        List<Long> numbers = new ArrayList<>();  
  
        try (BufferedReader br = new BufferedReader(new  
FileReader(filePath))) {  
            String line = br.readLine();  
            if (line != null) {  
                String[] parts = line.trim().split("\\s+");  
                for (String part : parts) {
```



```
        try {
            numbers.add(Long.parseLong(part));
        } catch (NumberFormatException e) {
            System.out.println("Skipping invalid number: " +
part);
        }
    }
}

List<Long> evenNumbers = getEven(numbers);
List<Long> oddNumbers = getOdd(numbers);

write(evenNumbers, "even.txt");
write(oddNumbers, "odd.txt");

        System.out.println("Odd and even numbers written
successfully.");

    } catch (IOException e) {
        System.out.println("Error reading file: " + e.getMessage());
    }
}

public List<Long> getEven(List<Long> numbers) {
    List<Long> even = new ArrayList<>();
    for (long num : numbers) {
        if (num % 2 == 0) {
            even.add(num);
        }
    }
    return even;
}

public List<Long> getOdd(List<Long> numbers) {
    List<Long> odd = new ArrayList<>();
    for (long num : numbers) {
        if (num % 2 != 0) {
            odd.add(num);
        }
    }
}
```

```

        return odd;
    }

    public void write(List<Long> numbers, String filePath) {
        try (BufferedWriter bw = new BufferedWriter(new
FileWriter(filePath))) {
            for (long num : numbers) {
                bw.write(num + " ");
            }
        } catch (IOException e) {
            System.out.println("Error writing to file " + filePath + ": "
+ e.getMessage());
        }
    }
}

```

```

Cannot load PSReadline module. Console is running without PSReadline.
PS E:\Java\Workshop\OED> & "C:\Program Files\Eclipse Adoptium\jdk-21.0.7.6-hotspot\bin\java.exe" "-XX:+ShowCodeDetailsInExceptionMessages" "-cp" "C:\Users\Vaibhu Shrestha\AppData\Roaming\Code\User\workspaceStorage\c320c77201fe0ff702f6e29bc34c3af\redhat.\java\jdk_ws\Wor
kshop\OED\src\main\resources\NumberSeparator"
Odd and even numbers written successfully.
PS E:\Java\Workshop\OED>

```

- Write a program that reads a text file and counts the number of words in it. Display the total count at the end.

```

import java.io.*;

public class WordCount {
    public static void main(String[] args) {
        String filePath = "filename.txt"; // Replace with your file path
        int wordCount = 0;

        try (BufferedReader br = new BufferedReader(new
FileReader(filePath))) {
            String line;

            while ((line = br.readLine()) != null) {
                // Split line into words based on whitespace
            }
        }
    }
}

```

```

        String[] words = line.trim().split("\\s+");
        if (words.length == 1 && words[0].isEmpty()) {
            // Empty line, no words
            continue;
        }
        wordCount += words.length;
    }

    System.out.println("Total number of words: " + wordCount);

} catch (IOException e) {
    System.out.println("Error reading file: " + e.getMessage());
}
}
}

```

```

PS E:\Jaca\Workshop\INE>
PS E:\Jaca\Workshop\INE> cd 'e:\Jaca\Workshop\INE'; & 'C:\Program Files\Eclipse Adoptium\jdk-21.0.7.6-hotspot\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\Vadju\Shrestha\AppData\Roaming\Code\User\workspaceStorage\c328cf7201fe9ff7026e29b6f4c8a1\redhat-java\src-as\Workshop\INE_f4cd9709\bin' 'WordCount'
Total number of words: 24
PS E:\Jaca\Workshop\INE>

```

10. Write a program to take the student information from the user and write the content to the csv file. The student class should have the following attributes.

- a. firstName, lastName, address, contactNumber, semester

```

import java.io.*;

import java.util.*;

class Student {
    String firstName;
    String lastName;
    String address;
    String contactNumber;
    int semester;
}

```

```
        public Student(String firstName, String lastName, String address,
String contactNumber, int semester) {
            this.firstName = firstName;
            this.lastName = lastName;
            this.address = address;
            this.contactNumber = contactNumber;
            this.semester = semester;
        }

        // Convert student data to CSV format line
        public String toCSV() {
            // Escape commas in address if needed
            return String.join(",", firstName, lastName, address.replace(",",
";"), contactNumber, String.valueOf(semester));
        }
    }

    public class StudentCSVWriter {
        public static void main(String[] args) {
            Scanner scanner = new Scanner(System.in);
            List<Student> students = new ArrayList<>();

            System.out.print("How many students do you want to enter? ");
            int n = Integer.parseInt(scanner.nextLine());

            for (int i = 0; i < n; i++) {
                System.out.println("\nEnter details for student " + (i + 1) +
":");

                System.out.print("First Name: ");
                String firstName = scanner.nextLine();

                System.out.print("Last Name: ");
                String lastName = scanner.nextLine();

                System.out.print("Address: ");
                String address = scanner.nextLine();

                System.out.print("Contact Number: ");
                String contactNumber = scanner.nextLine();
            }
        }
    }
}
```

```
        System.out.print("Semester: ");
        int semester = Integer.parseInt(scanner.nextLine());

        students.add(new Student(firstName, lastName, address,
contactNumber, semester));
    }

    writeToCSV(students, "students.csv");
    System.out.println("\nStudent data written to students.csv");
}

public static void writeToCSV(List<Student> students, String fileName)
{
    try (BufferedWriter bw = new BufferedWriter(new
FileWriter(fileName))) {
        // Write CSV header
        bw.write("FirstName,LastName,Address,ContactNumber,Semester");
        bw.newLine();

        // Write student data
        for (Student s : students) {
            bw.write(s.toCSV());
            bw.newLine();
        }

    } catch (IOException e) {
        System.out.println("Error writing to CSV file: " +
e.getMessage());
    }
}
```

```
Cannot load PSHeadline module. Console is running without PSHeadline.  
PS E:\Jaca\Workshop\IDE> & "C:\Program Files\Eclipse Adoptium\jdk-21.0.7.6-hotspot\bin\java.exe" ^  
-XX:+ShowCodeDetailsInExceptionMessages" ^-cp "C:\Users\Yajju Shrestha\AppData\Roaming\Code\User\workspaceStorage\c320cf7201fe0fff7702f6e29bc14c3ef\vscode\java\jdk_ws\Wor  
kshop\IDE_14c097d9\bin" "StudentCSWriter"  
How many students do you want to enter? 2  
  
Enter details for student 1:  
First Name: yajju chamsi  
Last Name: shrestha  
Address: dhapakhel  
Contact Number: 98458526154  
Semester: 3  
  
Enter details for student 2:  
First Name: roczy  
Last Name: goud  
Address: lalitpur  
Contact Number: 8946512346513  
Semester: 5  
  
Student data written to students.csv  
PS E:\Jaca\Workshop\IDE> []
```